Practical 13

Design a web page demonstrating different Core JavaScript references (Array, Boolean, Date, Function, Math, Number, Object, String, RegExp).

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Practical 13 - Core JavaScript References</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
            max-width: 800px;
            margin: auto;
        }

        .box {
            background-color: #eeeeee;
            padding: 15px;
            margin: 15px 0;
            border-radius: 8px;
        }

        button {
            padding: 10px 20px;
            cursor: pointer;
        }

        p {
            color: blue;
            font-weight: bold;
```

```
        }
    </style>
</head>

<body>

<section>

    <h1>Practical 13</h1>
    <h2>Core JavaScript References</h2>

    <div class="box">
        <h3>1. Array</h3>
        <button onclick="arrayExample()">Show Array</button>
        <p id="arrayResult"></p>
    </div>

    <div class="box">
        <h3>2. Boolean</h3>
        <button onclick="booleanExample()">Show Boolean</button>
        <p id="booleanResult"></p>
    </div>

    <div class="box">
        <h3>3. Date</h3>
        <button onclick="dateExample()">Show Date</button>
        <p id="dateResult"></p>
    </div>

    <div class="box">
        <h3>4. Function</h3>
        <button onclick="functionExample()">Call Function</button>
        <p id="functionResult"></p>
    </div>

    <div class="box">
        <h3>5. Math</h3>
        <button onclick="mathExample()">Show Math</button>
        <p id="mathResult"></p>
    </div>

    <div class="box">
```

```
        <h3>6. Number</h3>
        <button onclick="numberExample()">Show Number</button>
        <p id="numberResult"></p>
    </div>

    <div class="box">
        <h3>7. Object</h3>
        <button onclick="objectExample()">Show Object</button>
        <p id="objectResult"></p>
    </div>

    <div class="box">
        <h3>8. String</h3>
        <button onclick="stringExample()">Show String</button>
        <p id="stringResult"></p>
    </div>

    <div class="box">
        <h3>9. RegExp</h3>
        <button onclick="regexpExample()">Show RegExp</button>
        <p id="regexpResult"></p>
    </div>

</section>


<script>
// 1. ARRAY
function arrayExample() {

    let fruits = ["Apple", "Banana", "Mango"];

    document.getElementById("arrayResult").innerHTML =
        "Array = " + fruits.join(", ");
}

// 2. BOOLEAN
function booleanExample() {

    let value = true;

    document.getElementById("booleanResult").innerHTML =
```

```
        "Boolean Value = " + value;
}

// 3. DATE
function dateExample() {

    let today = new Date();

    document.getElementById("dateResult").innerHTML =
        "Current Date and Time = " + today;
}

// 4. FUNCTION
function greet() {

    return "Hello! This message is returned by a function.";
}

function functionExample() {

    document.getElementById("functionResult").innerHTML =
        greet();
}

// 5. MATH
function mathExample() {

    let number = 25;

    document.getElementById("mathResult").innerHTML =
        "Square Root of 25 = " + Math.sqrt(number);
}

// 6. NUMBER
function numberExample() {

    let number = 25.6789;

    document.getElementById("numberResult").innerHTML =
        "Number = " + number.toFixed(2);
}
```

```
// 7. OBJECT
function objectExample() {

    let student = {
        name: "Rahul",
        age: 20,
        course: "Computer Science"
    };

    document.getElementById("objectResult").innerHTML =
        "Name = " + student.name +
        ", Age = " + student.age +
        ", Course = " + student.course;
}

// 8. STRING
function stringExample() {

    let text = "JavaScript";

    document.getElementById("stringResult").innerHTML =
        "String = " + text +
        ", Length = " + text.length;
}

// 9. REGEXP
function regexpExample() {

    let pattern = /JavaScript/i;

    let text = "I am learning JavaScript.";

    let result = pattern.test(text);

    document.getElementById("regexpResult").innerHTML =
        "Pattern found = " + result;
}

</script>
</body>
</html>
```

Practical 12

Write Javascript code for

a. Finding Palindrome numbers

index.html

```
<!DOCTYPE html>
<html>

<head>
    <title>Practical 14(a) - Palindrome Number</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            max-width: 600px;
            margin: auto;
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
        }

        .box {
            background-color: #eeeeee;
            padding: 20px;
        }

        input,
        button {
            padding: 10px;
            margin: 5px;
        }

        #result {
            color: blue;
            font-weight: bold;
```

```
        }
    </style>
</head>

<body>
    <section>
        <h1>Practical 14(a)</h1>
        <h2>Finding Palindrome Number</h2>
        <div class="box">
            <label>Enter a number:</label>
            <input type="number" id="number">
            <button onclick="checkPalindrome()">
                Check Palindrome
            </button>
            <p id="result"></p>
        </div>
    </section>

    <script>
        function checkPalindrome() {
            let number = Number(document.getElementById("number").value);
            let original = number;
            let reverse = 0;
            while (number > 0) {
                let digit = number % 10;
                reverse = reverse * 10 + digit;
                number = Math.floor(number / 10);
            }

            if (original === reverse) {
                document.getElementById("result").innerHTML =
                    original + " is a Palindrome Number.";
            } else {
                document.getElementById("result").innerHTML =
                    original + " is not a Palindrome Number.";
            }
        }
    </script>

</body>
</html>
```

b. Calculating reverse of a number

index.html

```
<!DOCTYPE html>
<html>

<head>
    <title>Practical 14(b) - Reverse of Number</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            max-width: 600px;
            margin: auto;
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
        }

        .box {
            background-color: #eeeeee;
            padding: 20px;
        }

        input,
        button {
            padding: 10px;
            margin: 5px;
        }

        #result {
            color: blue;
            font-weight: bold;
        }
    </style>
</head>
```

```
<body>
    <section>
        <h1>Practical 14(b)</h1>
        <h2>Reverse of a Number</h2>
        <div class="box">
            <label>Enter a number:</label>
            <input type="number" id="number">
            <button onclick="reverseNumber()">
                Reverse
            </button>
            <p id="result"></p>
        </div>
    </section>
    <script>
        function reverseNumber() {
            let number = Number(document.getElementById("number").value);
            let reverse = 0;
            while (number > 0) {
                let digit = number % 10;
                reverse = reverse * 10 + digit;
                number = Math.floor(number / 10);
            }
            document.getElementById("result").innerHTML =
                "Reverse = " + reverse;
        }
    </script>
</body>
</html>
```

c. Displaying the following pattern

```
*
**
***
```

index.html

```
<!DOCTYPE html>
<html>

<head>
    <title>Practical 14(c) - Star Pattern</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            max-width: 600px;
            margin: auto;
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
        }

        .box {
            background-color: #eeeeee;
            padding: 20px;
        }

        button {
            padding: 10px 20px;
            cursor: pointer;
        }

        #result {
            color: blue;
            font-size: 25px;
            font-weight: bold;
```

```
        }
    </style>
</head>

<body>
    <section>
        <h1>Practical 14(c)</h1>
        <h2>Star Pattern</h2>
        <div class="box">
            <button onclick="displayPattern()">
                Display Pattern
            </button>
            <p id="result"></p>
        </div>
    </section>

    <script>
        function displayPattern() {
            let result = "";
            for (let i = 1; i <= 3; i++) {
                for (let k = 1; k <= 3 - i; k++) {
                    result += " ";
                }
                for (let j = 1; j <= i; j++) {
                    result += "* ";
                }
                result += "<br>";
            }
            document.getElementById("result").innerHTML = result;
        }
    </script>

</body>
</html>
```

d. Displaying the following pattern

```
*
**
****
******
```

index.html

```
<!DOCTYPE html>
<html>

<head>
    <title>Practical 14(d) - Star Pattern</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            max-width: 600px;
            margin: auto;
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
        }

        .box {
            background-color: #eeeeee;
            padding: 20px;
        }

        button {
            padding: 10px 20px;
            cursor: pointer;
        }

        #result {
            color: blue;
            font-size: 25px;
```

```
            font-weight: bold;
        }
    </style>
</head>

<body>
    <section>
        <h1>Practical 14(d)</h1>
        <h2>Star Pattern</h2>
        <div class="box">
            <button onclick="displayPattern()">
                Display Pattern
            </button>
            <p id="result"></p>
        </div>
    </section>

    <script>
        function displayPattern() {
            let result = "";
            let stars = [1, 2, 4, 6];
            for (let i = 0; i < stars.length; i++) {
                for (let k = 0; k < (6 - stars[i]); k++) {
                    result += " ";
                }
                for (let j = 1; j <= stars[i]; j++) {
                    result += "* ";
                }
                result += "<br>";
            }
            document.getElementById("result").innerHTML = result;
        }
    </script>

</body>
</html>
```